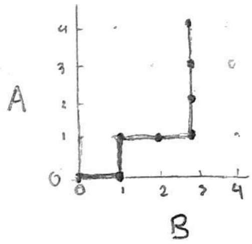
Problem 3:

a) Modeling Ω_1: Let the votes be indexed according to the order in which they are counted, i.e., a vote $v \in \{1, \dots, a+b\}$. An outcome $M \in \Omega_1$ keeps track of the indexes where candidate A received a vote, s.t., $M \subset \{1, \dots, a+b\}$ and $|M| = a$. Since $a+b$ is the total number of votes, and candidate can win one indexed vote only once (i.e., w/o replacement) then $|\Omega_1| = \binom{a+b}{a}$. Also, $\mathcal{F}_1 = P(\Omega_1)$ and $\mathbb{P}_1(F) = \frac{|F|}{|\Omega_1|}$, $F \subset \mathcal{F}_1$.

Modeling Ω_2: Let $i \in \{0, \dots, a+b\}$ be the number of votes counted at any given point during the counting process. An outcome $g \in \Omega_2$ is a function from i to a tuple (b_i, a_i) containing the distribution of votes between candidates B and A (respectively) when i votes have been counted (s.t., $i = b_i + a_i$). Since there is only one more vote counted between $i-1$ and i (and assuming all votes are valid) then the vote is either given to A, so that $a_i = a_{i-1} + 1 \Leftrightarrow a_{i-1} = a_i - 1$ and $b_i = b_{i-1}$ and then $g(i) - g(i-1) = (b_i, a_i) - (b_i, a_i - 1) = (0,1)$, or given to B, so that $g(i) - g(i-1) = (1,0)$. Thus, g can be interpreted as a path in a grid going righ if B wins and up if A wins, e.g., the plot on the left represents the map of a path in which B obtains the first vote (right) A the second (up) then B obtains 2 votes and finally A obtains 3 votes winning the election with 4 votes against 3.



Each map consists of $a+b$ steps, of which a must be "up" (an the remaining b must be "right"). Since a step can only be taken once (otherwise it would not be a path) then it is equivalent to draw without replacement and without order, therefore $|\Omega_2| = \binom{a+b}{a}$, $\mathcal{F}(\Omega_2) = P(\Omega)$ and $\mathbb{P}(F) = \frac{|F|}{|\Omega_2|}$, $F \in \mathcal{F}$.

b) $A_1 = \{ g \in \Omega_2 \mid g(1) = (1,0) \}$

$A_2 = \{ g \in \Omega_2 \mid g(1) = (0,1) \wedge \exists k \in \{2, \dots, a+b\} : b_k = a_k \text{ where } g(k) = (b_k, a_k) \}$

(note that $g(0)$ is not considered ($i > 0$), as the counting has not started).

Proof $A_1 \cap A_2 = \emptyset$: Since $g \in \Omega_2$ cannot simultaneously satisfy $g(1) = (1,0)$ and $g(1) = (0,1)$, then $g \in A_1 \Rightarrow g \notin A_2$ and $g \in A_2 \Rightarrow g \notin A_1$ and, therefore, $A_1 \cap A_2 = \emptyset$ $\square$

Proof $|A_1| = |A_2|$: We prove that there is a bijection by showing:

(i) $\forall g \in A_1, \exists g' \in A_2$: Let $g \in A_1$ s.t. $g(i) = (b_i, a_i)$. Since $g(0) = (0,0)$ and $g(1) = (1,0)$, g begins below the diagonal. Since the final point is (b, a) and $a > b$, the path ends above the diagonal, so it intersects the diagonal at least once. Let $k \in \{2, \dots, a+b\}$ be the first index where $b_k = a_k$ and let

$$g'(i) = \begin{cases} (a_i, b_i) & \text{for } 0 \leq i \leq k \\ (b_i, a_i) & \text{for } k < i \leq a+b \end{cases} \quad \text{(the path is reflected)}$$

Then, $g'(0,0) = 0$, $g'(a+b) = (b,a)$, $g'(i) - g'(i-1) \in \{(0,1),(1,0)\}$ and $g' \in \Omega_2$. Also, $g'(1) = (0,1)$ and $a_k = b_k$. Then, $g' \in A_2$.

(ii) $\forall g \in A_2, \exists g' \in A_1$: Let $g \in A_2$ s.t. $g(i) = (b_i, a_i)$. Let $k \in \{2, \dots, a+b\}$ be the first index where $b_k = a_k$ (which, by definition, it exists) and let:

$$g'(i) = \begin{cases} (a_i, b_i) & \text{for } 0 \leq i \leq k \\ (b_i, a_i) & \text{for } k < i \leq a+b \end{cases} \quad \text{(the path is reflected)}$$

Then, $g'(0) = (0,0)$, $g'(a+b) = (b,a)$ and $g'(i) - g'(i-1) \in \{(0,1),(1,0)\}$ which implies that $g' \in \Omega_2$. Since $g'(1) = (1,0)$ (as $g(1) = (0,1)$) and $a_k = b_k$, then $g' \in A_1$

Since $g'' = g$ (by (i)), i.e., a bijection, $|A_1| = |A_2|$ $\square$

c) Let E be the event of the candidate A strictly winning during the entire counting, i.e., E contains all paths that lie above the diagonal. Formally:

$$E = \{g \in \Omega_2 \mid a_i > b_i \;\; \forall i \in \{1, \dots, a+b\}\}$$

The events in A_1 contains the paths that fail the condition $a_i > b_i$ for $i = 1$ (as $g(1) = (1,0)$, $g \in A_1$), an A_2 contains the paths that succeed the condition $a_i > b_i$ (as $g(1) = (0,1)$, $g \in A_2$) but fail it at some later point because the path hits the diagonal ($\exists_i \in \{2, \dots, a+b\} : b_i = a_i$). Therefore, $A_1 \cup A_2$ contains all paths where $b_i \geq a_i$ at least once ($\exists_i \in \{1\} \cup \{2, \dots, a+b\} : b_i \geq a_i$). Then, $(A_1 \cup A_2)^c$ contains all path where $b_i \geq a_i$ is never true:

$$\nexists i \in \{1, \dots, a+b\} : b_i \geq a_i \Rightarrow \forall_i \in \{1, \dots, a+b\} : b_i > a_i,$$

which fulfills the definition of E, i.e., $E = (A_1 \cup A_2)^c$. We then have:

$|A_1 \cup A_2| \overset{A_1 \cap A_2 = \emptyset}{=} |A_1| + |A_2| \overset{|A_1| = |A_2|}{=} 2|A_1|$. For A_1 the first step is the only one that is fixed ($g(1) = (1,0)$), leaving a "up" steps and $b-1$ "right" steps implying $|A_1| = \binom{a+b-1}{a}$. Putting it all together:

$$\mathbb{P}(E) = \mathbb{P}((A_1 \cup A_2)^c) = \mathbb{P}(\Omega_2 \setminus (A_1 \cup A_2)) = 1 - \mathbb{P}(A_1 \cup A_2) = 1 - \frac{|A_1 \cup A_2|}{|\Omega_2|} = 1 - \frac{2\binom{a+b-1}{a}}{\binom{a+b}{a}}$$

$$= 1 - 2\frac{\frac{(a+b-1)!}{a!\,(b-1)!}}{\frac{(a+b)!}{a!\,b!}} = 1 - \frac{2\cancel{(a+b-1)!}\, b\, \cancel{(b-1)!}}{(a+b)\cancel{(a+b-1)!}\,\cancel{(b-1)!}} = 1 - \frac{2b}{a+b} = \frac{a-b}{a+b} \quad \square$$